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//Program to take multiple numbers as input from user and do different functions
#include<stdio.h>

int n;float sum=0,avg=0;
int c;
float arr[50];

void main(){
        input();
        output();
        while(1){

        printf("\n\nWhat do you want to do?\n\n1.Average\n2.ascending sort\n3.descending sort\n4.Search for a number\n\n<Any other button to exit> \n");
        scanf("%d",&c);
        switch(c){
        case 1:average();break;
        case 2:sort();break;
        case 3:descend_sort();break;
        case 4:search();break;

        default:printf("\nexiting the program"); exit(0);break;
      }
    }
}
void input(){
        printf("How many numbers :");
        scanf("%d",&n);
        printf("\nGive %d float numbers :",n);
```

```
		for(int i=0;i<n;i++){
		scanf("%f",&arr[i]);
		}
}
void output(){
		printf("\nThe %d float numbers are:    ",n);
		for(int i=0;i<n;i++){
		printf("%.2f    ",arr[i]);
		}
}
void average(){
		sum=0;
		for(int j=0;j<n;j++){
			sum=sum+arr[j];
		}
		avg=sum/n;
		printf("\nAverage: %.2f",avg);

}
void sort(){
		float temp;
		for(int j=0;j<n;j++){
			for(int i=0;i<n;i++){
				if(arr[j]<arr[i]){
					temp=arr[j];
					arr[j]=arr[i];
					arr[i]=temp;
				}
			}
```

```
        }
        printf("\nAscending order : ");
        for(int i=0;i<n;i++){
            printf("%.2f    ",arr[i]);
        }

}
void descend_sort(){
        float tempt;
        for(int j=0;j<n;j++){
            for(int i=0;i<n;i++){
                if(arr[j]>arr[i]){
                    tempt=arr[j];
                    arr[j]=arr[i];
                    arr[i]=tempt;
                }
            }
        }
        printf("\nDescending order : ");
        for(int i=0;i<n;i++){
            printf("%.2f    ",arr[i]);
        }

}

void search(){
        float s;
        int count=0;
        int found=1;
```

```
            printf("\nWhich number do you want to search: ");
            scanf("%f",&s);
            for(int i=0;i<n;i++){
            if(arr[i]==s){
                  count=1;
                  printf("\nFound %d times",found);
                  found++;
            }
      }
            if(!count){
                  printf("\nNot found");
            }
}
```

Output :

How many numbers :5

Give 5 float numbers :2.5

6

1.05

6

7.8

The 5 float numbers are: 2.50 6.00 1.05 6.00 7.80

What do you want to do?

1.Average

2.ascending sort

3.descending sort

4.Search for a number

<Any other button to exit>

1

Average: 4.67

What do you want to do?

1.Average

2.ascending sort

3.descending sort

4.Search for a number

<Any other button to exit>

2

Ascending order : 1.05 2.50 6.00 6.00 7.80

What do you want to do?

1.Average

2.ascending sort

3.descending sort

4.Search for a number

<Any other button to exit>

3

Descending order : 7.80 6.00 6.00 2.50 1.05

What do you want to do?

1.Average

2.ascending sort

3.descending sort

4.Search for a number

<Any other button to exit>

4

Which number do you want to search: 2.5

Found 1 times

What do you want to do?

1.Average

2.ascending sort

3.descending sort

4.Search for a number

<Any other button to exit>

9

exiting the program

Process returned 0 (0x0) execution time : 48.443 s